

CASE STUDY - ASSESSMENT TOOL 2

Case Study 1: Warehouse Robot Using Reinforcement Learning

Problem

A logistics company wants a robot to pick, pack, and transport items automatically. The robot should avoid obstacles and complete tasks quickly.

MDP Components

States

- Robot location
- Item location
- Battery level
- Obstacle position
- Carrying item or not

Actions

- Move forward
- Turn left/right
- Pick item
- Drop item
- Recharge battery

Rewards

- Pick item: **+10**
- Deliver item: **+20**
- Reach destination quickly: **+5**

- Hit obstacle: **-20**
- Extra movement: **-1**

Learning

The robot learns the shortest and safest path.

Outcome

- Faster delivery
 - Fewer accidents
 - Better warehouse efficiency
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Case Study 2: Movie Recommendation System

Problem

A streaming platform wants to recommend movies based on users' interests and watch history.

MDP Components

States

- User's favorite genre
- Movies watched
- Ratings
- Time of day

Actions

- Recommend Action movie
- Recommend Comedy
- Recommend Drama
- Recommend Horror

Rewards

- User watches movie: **+10**
- User likes movie: **+5**

- User skips movie: **-3**
- User stops watching early: **-5**

Learning

The system learns which movies each user likes.

Outcome

- Better recommendations
 - More watch time
 - Higher user satisfaction
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Case Study 3: Smart Agriculture System

Problem

A farm uses RL to decide how much water and fertilizer crops need.

MDP Components

States

- Soil moisture
- Weather
- Crop growth stage
- Temperature

Actions

- Add water
- Add fertilizer
- Do nothing

Rewards

- Healthy crop growth: **+20**
- High crop yield: **+50**
- Waste water: **-10**

- Crop damage: **-20**

Learning

The system learns the best watering and fertilizer schedule.

Outcome

- Higher crop yield
 - Less water usage
 - Lower fertilizer cost
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Case Study 4: Cybersecurity System

Problem

A company wants an RL system to detect cyber attacks and respond automatically.

MDP Components

States

- Network traffic
- Login attempts
- Threat level
- Firewall status

Actions

- Allow traffic
- Block IP address
- Alert administrator
- Quarantine device

Rewards

- Detect attack: **+20**
- Stop attack: **+50**
- False alarm: **-10**

- Miss attack: **-50**

Learning

The system learns to identify attacks quickly.

Outcome

- Better security
 - Faster attack detection
 - Fewer false alarms
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Case Study 5: Ride-Sharing Company

Problem

A ride-sharing company wants to assign drivers and set ride prices automatically.

MDP Components

States

- Number of drivers
- Number of passengers
- Traffic
- Weather
- Time of day

Actions

- Assign driver
- Increase price
- Decrease price
- Move driver to another area

Rewards

- Successful ride: **+20**
- Happy customer: **+10**

- Ride cancelled: **-10**
- Driver waiting too long: **-5**

Learning

The system learns the best way to assign drivers and adjust prices.

Outcome

- More completed rides
- Shorter waiting time
- Better driver utilization
- Higher customer satisfaction